Experiment 3: Only the Engine with Two People on an Incline

The third experiment on the same day was conducted on an ascending path with an inclination of 1 in 60 [18] and steam at 30 lbs. It had no carriages attached, just the steam engine with two persons. This had the reduced total weight of 3,200 lbs, and it attained a speed of 3.5 miles per hour.

The report stated that in these experiments, the total force exerted at the end of the arm was much less than what was calculated. This was owing to defects in the furnace and the chimney, which produced an insufficient draught. Hence it was impossible to keep up a proper fire. These defects were soon remedied and the power subsequenty increased—finally sufficient to propel the engine at an increased speed.

EXPERIMENTS LEAD TO IMPROVIZATION!

The next set of experiments whose records are available were conducted on Wednesday, 12 September 1838. These were for a heavier engine and train, with eight passengers, at 4.5 miles per hour. The technical specifications were a bit altered, with the engine, the boiler and the carriage weighing 3,000 lbs (as against the earlier 2,800 lbs), the tender or coal car and coal weighing 1,600 lbs, and a carriage with eight passengers weighing 2,400 lbs —weighing a total of 7,000 lbs, or 3.12 tonnes. The pressure in the boiler was about 50 lbs. The whole train moved at the rate of about 4½ miles per hour, the highest speed achieved so far. The road where this experiment was conducted was not very level; there was an ascent of about one foot in a thousand.

It is known from the paper about the locomotives that experiments ceased abruptly after the second test, as Captain Cotton, the driving force behind the experiments and the railway, became ill.

Smith writes:

> …It is much to be regretted that abrupt termination of experiments by the sudden departure of their indefatigable author, owing to ill health, has deprived us of the opportunity of becoming acquainted with the results of a further prosecution of the enquiry in the capabilities of this machine. This is the more unfortunate, since the most important question relative to it, viz. its comparative efficiency as a mechanical agent with the other means of applying steam now in use, rests upon the determination of a point still left undecided. [19]